SOFTWARE ENGINEERING PROJECT

Prometheus:
Computer Vision-Based Gym Machine Occupancy Detection and Spatial Analytics

BY

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Abstract

Prometheus is an integrated AI system designed to bridge the gap between gym infrastructure and the member experience by repurposing existing CCTV networks into real-time analytical engines. Traditional gym management relies on manual tracking or expensive hardware sensors, both of which fail to provide real-time, scalable data. This project implements a Computer Vision pipeline using the YOLO (You Only Look Once) architecture to detect equipment occupancy and analyze spatial traffic patterns. The system provides a dual-interface solution: an iOS application for members to check live machine availability and a web-based dashboard for owners to visualize facility utilization via heatmaps. By leveraging Edge Computing to ensure privacy and low latency, Prometheus optimizes workout efficiency for users and provides data-driven insights for facility operators to improve equipment Return on Investment (ROI).

Acknowledgement

We would like to express our deepest gratitude to our project advisor and the Data Analysis and Knowledge Discovery Lab (DAKDL) for providing the technical guidance and resources necessary to conduct this research. We are also grateful to Kasetsart University and the staff at the Kasetsart Engineering Fit and Firm (KEFF) gym for their support in data collection and environment testing. Finally, we thank our peers and families for their encouragement throughout the development of this project.

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Chapter 1

Introduction

1.1 Background

The fitness industry has seen a significant surge in membership, yet the technological infrastructure within gyms remains largely stagnant. Most facilities still rely on manual equipment logging or basic membership check-ins. For members, the lack of real-time information regarding equipment availability leads to inefficient workouts and "availability uncertainty." For owners, the absence of granular data on how floor space is utilized results in suboptimal equipment purchasing and layout decisions. With the advancement of Computer Vision and the ubiquity of CCTV systems, there is a significant opportunity to transform passive video feeds into actionable spatial intelligence.

1.2 Problem Statement

Current gym operations suffer from two primary inefficiencies:

1. Member Frustration: Users often find their planned equipment occupied upon arrival, leading to routine disruption and decreased facility satisfaction.

2. Operational Blind Spots: Gym owners lack objective data on equipment utilization rates and "dead zones," forcing them to rely on guesswork for capital expenditure and floor plan optimization.

3. High Cost of Alternatives: Existing IoT sensor solutions are prohibitively expensive to install and maintain on every individual machine in a high-traffic environment.

1.3 Solution Overview

Prometheus solves these issues by implementing a non-intrusive AI system that utilizes existing gym CCTV cameras. The system uses an Edge-to-Cloud architecture where local processing units analyze video streams to identify machine states (Free vs. Occupied). This metadata is then synchronized to a cloud database, powering a real-time mobile tracker for members and a spatial analytics dashboard for management.

1.3.1 Features

1. **Real-time Occupancy Tracker:** A live mobile interface showing which machines are currently in use or available to help members plan their sets.

2. **Spatial Heatmaps:** A web-based visual representation of high-traffic areas and underutilized equipment for facility owners.

3. **AI Diet Estimator:** A database-driven tool that calculates nutritional needs and recommends food items based on user profiles.

4. **Progress Logger:** A feature for members to record sets, reps, and weights, integrated with visual progress charts and historical data.

5. **Workout Routine Generator:** An intelligent routine creator tailored to user goals, physical profile, and real-time equipment availability.

6. **Automated Utilization Reports:** Weekly data summaries for owners regarding peak hours, machine popularity, and equipment ROI.

1.4 Target User

The Prometheus system is designed for two distinct demographics, with a primary focus on facility management and business intelligence:

- **Primary Users (Gym Owners and Facility Managers):** These are the central stakeholders who utilize the system to drive operational efficiency. They require high-level spatial analytics to identify equipment usage patterns, minimize "dead zones," and optimize facility layouts. By leveraging the web-based dashboard, these users transform raw computer vision data into actionable business insights to improve equipment Return on Investment (ROI) and long-term member retention.
- **Secondary Users (Gym-goers):** Individual patrons who benefit from the downstream data generated by the system. These users range from beginners to advanced athletes who utilize the mobile interface to minimize wait times and follow AI-generated routines. While they are frequent interactors with the platform, their experience is an extension of the primary analytical framework established for the facility.

1.5 Terminology

- **Computer Vision (CV):** A field of Artificial Intelligence that enables computers to derive meaningful information from digital images or videos.
- **YOLO (You Only Look Once):** A state-of-the-art, real-time object detection system used by Prometheus for identifying equipment and patrons.
- **Edge Computing:** Processing data near the source (the camera) to reduce latency and enhance privacy by keeping raw video footage local.
- **Region of Interest (ROI):** Specifically defined coordinates within a video frame that correspond to individual pieces of gym equipment.

- **Occupancy Logic:** The algorithmic determination of a "busy" state based on the spatial intersection of "Person" and "Machine" detection boxes.

Chapter 2

Literature Review and Related Work

2.1 Competitor Analysis

The competitive landscape for fitness technology is currently divided between manual tracking applications and high-cost industrial hardware. Prometheus positions itself as an automated, non-intrusive solution that leverages existing infrastructure to provide high-utility data at a lower entry cost for facility owners.

The following table provides a detailed comparison between Prometheus and existing solutions in the market, highlighting the gaps in real-time spatial intelligence.

Feature	**Prometheus**	**Manual Loggers**	**IoT Sensors**	**Gym Dashboards**
Real-time Occupancy	Automated	None	Automated	Manual Entry
Installation Cost	Low (CCTV)	Zero	High (Per-unit)	Moderate
Member Utility	High	High	Low	Moderate
Spatial Analytics	Heatmaps	None	Numeric Only	Attendance Only
Privacy	Edge-based	High	High	Moderate

Table 2.1: Comparison Table of Competitive Fitness Solutions

2.2 Literature Review

To build a technically sound and feasible system, this project draws from several key academic works and technical advancements in the fields of Computer Vision and Distributed Systems.

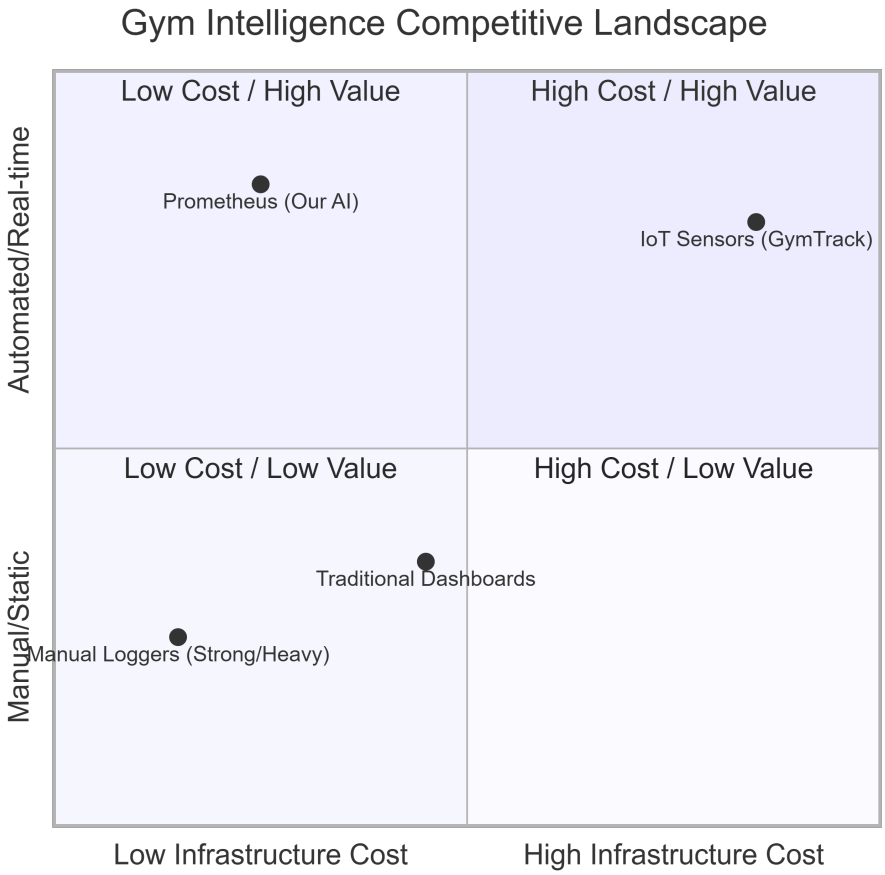

Figure 2.1: Competitive Landscape: Automation vs. Infrastructure Cost

2.2.1 Real-Time Object Detection (YOLO Architecture)

The foundation of the occupancy detection engine is based on the "You Only Look Once" (YOLO) architecture. Research by Redmon et al. established that frame-rate inference is possible by treating detection as a single regression problem. This project specifically evaluates the use of YOLOv8 and YOLOv10. The selection of YOLOv10 informs our choice for edge deployment, as its elimination of Non-Maximum Suppression (NMS) significantly reduces computational latency on local hardware.

2.2.2 Vision-Based Gym Monitoring (GymCam)

The "GymCam" project by Khurana et al. demonstrated the viability of using unconstrained camera feeds to track multiple users and exercises simultaneously. Their research indicates that while fine-grained action recognition is complex, identifying machine occupancy through spatial overlap is highly reliable. Prometheus extends this logic by utilizing a multi-camera RTSP network to provide floor-wide spatial analytics.

2.2.3 Large-Scale Multi-View Datasets (M3GYM)

For model training, this project utilizes the M3GYM dataset. Unlike standard datasets, M3GYM provides multi-view labels in real-world gym environments characterized by high occlusion. This informs the design of our "Occupancy Logic," allowing the system to distinguish between a person walking near equipment and a person actively utilizing a machine.

2.2.4 Privacy-Preserving Edge Computing

A primary constraint for gym-based AI is member privacy (GDPR/PDPA). Current literature suggests that performing inference at the "edge" allows raw video data to be discarded immediately after metadata extraction. This informs our choice of an Edge Processing Layer (e.g., ASUS NUC), ensuring that no identifiable video frames are ever transmitted to the cloud.

Chapter 3

Requirement Analysis

3.1 Stakeholder Analysis

The Prometheus project involves multiple parties that influence or are affected by the system's deployment. Identifying these stakeholders ensures that both the technical AI requirements and the business-end objectives are met.

- **Gym Members (Primary Users):** Individual athletes and gym patrons who utilize the facility. They are the primary consumers of the real-time occupancy data and progress tracking features. Their main interest is minimizing workout friction and maximizing time efficiency.

- **Gym Owners and Facility Managers (Secondary Users):** Business operators who require data-driven insights to optimize equipment placement and facility layout. They utilize the web dashboard for spatial analytics and ROI assessments.

- **Product Owner:** The individual responsible for the strategic vision of the Prometheus ecosystem. They ensure that the system remains compliant with data privacy regulations (PDPA/GDPR) and that the AI capabilities align with the intended problem domain.

- **Data Science/AI Developers:** Technical stakeholders responsible for the development, training, and maintenance of the YOLO models and the edge-to-cloud pipeline.

3.2 User Stories and Use Cases

3.2.1 User Stories

The following user stories define the core functional requirements from the perspective of the key stakeholders:

- **Gym Member:** "As a gym member, I want to see real-time machine occupancy via my mobile app so that I can plan my sets without waiting for equipment."
- **Gym Member:** "As a gym member, I want to receive an AI-generated workout routine based on currently available machines so that I can have a continuous, uninterrupted workout session."
- **Gym Owner:** "As a gym owner, I want to view a spatial heatmap of my facility so that I can identify 'dead zones' and relocate underutilized equipment to maximize my floor space value."
- **Gym Owner:** "As a gym owner, I want to access historical utilization reports so that I can make informed decisions about purchasing new equipment based on actual demand."
- **Product Owner:** "As a product owner, I want to ensure all raw video data is processed locally at the edge so that the system maintains member privacy and remains compliant with regional data protection laws."

3.2.2 Use Cases

The following use cases detail the interaction between the users and the Prometheus system.

Use Case 1: View Real-Time Machine Occupancy (Brief)

Actor: Gym Member
Goal: To determine if specific equipment is available before moving to that station.

Description: The member opens the Prometheus iOS application. The system retrieves the current occupancy metadata from the cloud database and displays a visual map of the gym floor. Machines are color-coded (e.g., Green for Free, Red for Occupied). The member identifies an available bench and proceeds with their workout.

Use Case 2: Analyze Facility Utilization Trends (Casual)

Actor: Gym Owner
Goal: To identify peak usage times and equipment popularity for business optimization.
Main Success Scenario: The owner logs into the web dashboard and selects a date range for analysis. The system aggregates the historical metadata and generates a spatial heatmap and a line graph showing utilization trends. The owner observes that the power racks are consistently at 100% capacity during 5:00 PM - 8:00 PM, while the stationary bikes are underutilized. Based on this, the owner decides to replace two bikes with an additional power rack.

Use Case 3: Detect and Update Machine State (Fully-Dressed)

Primary Actor: Edge Processing Unit (System Actor)
Stakeholders and Interests:

- **Gym Member:** Needs accurate, low-latency updates to avoid "false availability" readings.
- **Gym Owner:** Needs reliable data for long-term analytics.

Preconditions: The CCTV cameras are active and streaming via RTSP to the local edge hardware. The YOLO model is loaded and initialized.
Main Success Scenario:

1. The Edge Processing Unit receives a video frame from the gym floor.
2. The YOLO model detects a "Person" class and a "Bench Press" class within the defined Region of Interest (ROI).

3. The occupancy logic detects a significant spatial overlap between the person and the machine.
4. The system monitors this overlap for a continuous 5-second buffer (Temporal Smoothing).
5. The Edge Unit extracts the "Occupied" state and the Machine ID, then discards the raw video frame.
6. The metadata is transmitted to the cloud database.
7. The cloud database triggers a WebSocket update to all active iOS clients, turning the machine icon red.

Extensions (Alternative Flows):

- **3a. Occlusion detected:** If a person walks between the camera and the machine without using it, the temporal buffer (Step 4) prevents a false "Occupied" trigger.
- **5a. Network Failure:** If the edge unit cannot reach the cloud, it stores the metadata locally and syncs once the connection is restored to ensure analytics accuracy.

3.3 Use Case Diagrams

The Prometheus system utilizes a dual-interface approach to serve its distinct user groups. The following diagrams and descriptions outline the functional requirements for both Gym Members and Gym Owners, as well as the automated interactions performed by the system itself.

3.3.1 Gym Member Use Cases

The member-facing use cases focus on personal workout efficiency, progress tracking, and data privacy.

Figure 3.1: Use Case Diagram for Gym Members

- **UC-1: View Real-Time Machine Status:** Allows members to view a live map of the gym floor where equipment is color-coded based on availability (Green for Free, Red for Occupied)[cite: 17].
- **UC-2: Subscribe to Vacancy Notifications:** Users can select specific machines or exercises; the system sends a push notification the moment the required equipment becomes vacant[cite: 18].
- **UC-3: Log Workout Progress:** A standard interface for logging weights, sets, and reps, which are then visualized using progress charts[cite: 14].
- **UC-4: Generate AI Workout Routine:** Uses an LLM or structured logic to create a personalized plan based on the user's physical profile and goals[cite: 16].
- **UC-5: Use Diet Estimator:** A tool that recommends food items and calculates nutritional needs based on the user's specific fitness objectives[cite: 15].
- **UC-6: Manage Privacy & Data Rights:** Provides members with the ability to exercise their rights under the PDPA, such as requesting a copy of their data or the deletion of their personal logs[cite: 31].

3.3.2 Gym Owner Use Cases

The owner-facing use cases prioritize spatial intelligence, facility optimization, and business analytics.

Figure 3.2: Use Case Diagram for Gym Owners

- **UC-7: Analyze Machine Traffic:** Provides a real-time overview of equipment usage states across the facility floor[cite: 20].
- **UC-8: View Spatial Heatmaps:** Generates visual heatmaps to identify high-traffic hotspots and underutilized "dead zones"[cite: 23].
- **UC-9: Generate Utilization Reports:** Accesses a historical database to produce daily, weekly, or monthly reports on equipment popularity and peak usage times[cite: 22].

- **UC-10: Optimize Facility Layout:** Enables owners to make data-backed decisions on relocating assets to improve floor flow and utilization[cite: 23].
- **UC-11: Manage Operational Schedules:** Helps owners optimize staffing, cleaning, and maintenance schedules based on real-time and historical traffic volume[cite: 25].

3.3.3 System Use Case

This automated use case represents the core AI functionality that enables the user-facing features.

- **UC-12: Detect and Update Machine State:** An automated process where the Edge Processing Unit analyzes CCTV streams, identifies occupancy via intersection logic, and transmits anonymized metadata to the cloud while discarding raw video frames[cite: 31, 33].

3.4 User Interface Design

<TIP: Put the initial design of your application here. You can showcase a detailed design of a specific page or a sitemap of your application. See an example below./>

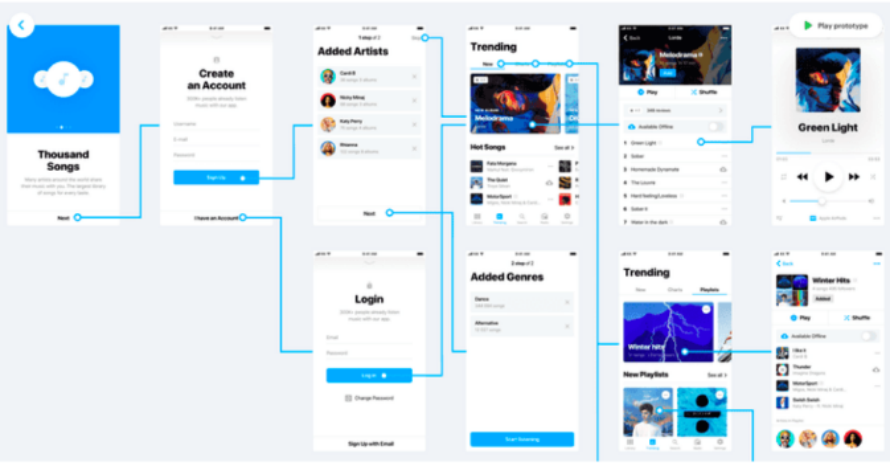

Figure 3.3: User Interface Design

Chapter 4

Software Architecture Design

4.1 System Architecture Overview

The Prometheus system follows a distributed Edge-to-Cloud architecture designed to balance real-time performance with data scalability. The architecture is composed of four primary layers: the Data Ingestion Layer (IP Cameras), the Edge Processing Layer (Inference Engine), the Cloud Service Layer (Data Management), and the Presentation Layer (Client Applications).

The system begins with 5 HD IP cameras streaming high-definition video via RTSP to a local Edge Processing Unit (e.g., ASUS NUC). This unit performs real-time object detection and state classification, immediately discarding raw video frames to ensure member privacy. The resulting anonymized metadata is transmitted to a PostgreSQL-backed cloud database, which serves as the central repository for historical utilization data. Finally, the Presentation Layer delivers live availability updates to the iOS application and spatial heatmaps to the PC Web Dashboard.

4.2 Software Design

The runtime behavior of Prometheus is driven by a continuous inference loop. When a member interacts with a machine, the system must detect, validate, and broadcast this state change within seconds.

Primary Sequence: Occupancy Update

1. The Edge Engine retrieves a frame and identifies "Person" and "Machine" objects using the YOLO model.

2. The system applies a **Spatial Overlap Algorithm** to determine if a person is actively using a machine. This logic calculates the intersection of the person's bounding box with a pre-defined Region of Interest (ROI) for that specific machine.

3. To prevent "flickering" from momentary occlusions or passersby, the system implements a **Temporal Smoothing** buffer: a state is only changed to "Occupied" if the overlap persists for at least 5 consecutive seconds.

4. Once validated, the status is pushed to the cloud, triggering a WebSocket event that updates the machine's color on all active iOS client maps.

4.3 AI and Data Design

4.3.1 Data Sources and Collection

The Prometheus AI models are developed using a hybrid data strategy to ensure both general robustness and facility-specific accuracy:

- **Public Benchmarks:** We utilize the M3GYM and QEVD datasets for baseline training. These sources provide millions of frames of traditional weightlifting and machine-based exercises from multiple CCTV-style angles.

- **Self-Collected Data:** To fine-tune the model for the specific layout of local facilities (such as the KEFF gym), we collect approximately 2,000 frames of facility-specific equipment.

- **Data Cleaning:** All training data undergoes perspective correction and manual annotation of "Free" vs. "Occupied" states. For privacy compliance, all human faces in the training set are blurred before being fed into the training pipeline.

4.3.2 Model Design

The core of the Prometheus intelligence is based on the **YOLO (You Only Look Once)** architecture. We have selected **YOLOv10** as the primary model for deployment on the ASUS NUC edge hardware.

- **Reasoning:** YOLOv10 was chosen over traditional R-CNN models because it offers the highest inference speed (frames per second) while maintaining the precision necessary for occupancy detection. Furthermore, YOLOv10's architecture eliminates Non-Maximum Suppression (NMS) overhead, which is critical for running multiple camera streams on a single edge device.
- **Integration:** The model is integrated into the system via an ONNX or TensorRT execution provider to maximize the utilization of the edge unit's GPU/NPU.

4.3.3 Evaluation

To ensure the system meets the established success criteria, the AI components are evaluated against the following metrics:

- **Mean Average Precision (mAP):** To measure the model's accuracy in identifying machines and people.
- **State Accuracy:** The percentage of correctly identified "Free" vs. "Occupied" states, with a target of $\geq 95\%$.
- **End-to-End Latency:** The time elapsed from a person sitting on a machine to the iOS app reflecting the "Occupied" status. Target: < 5 seconds.
- **False Positive Rate:** The frequency at which the system incorrectly marks a machine as occupied due to environmental noise or passersby.

Chapter 5

Software Development

5.1 Software Development Methodology

<TIP: Describe your software development methodology in this section. />

5.2 Technology Stack

<TIP: Describe your technology stack here. See the following example from ThaiProgrammer.org />

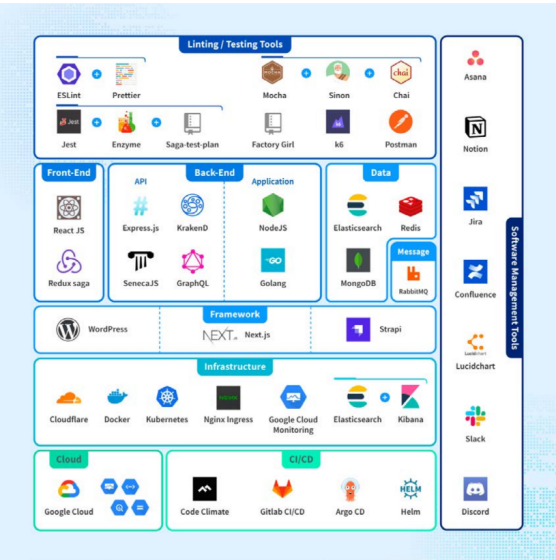

Figure 5.1: Example technology stack

5.3 Coding Standards

<TIP: Describe your coding standard for this project here. />

5.4 Schedule

<TIP: Show the schedule of your project. Use a Gantt chart. The items should agree with features you listed in Chapter 1. />

5.5 Progress Tracking Report

<TIP: Show that you have been working on this project overtime. It can be in the form of a burndown chart or a contribution graph from GitHub./>

Chapter 6

Deliverables

6.1 Software Solution

<TIP: Share a link to your Github repository. Showcase screenshots of the application and briefly describe each page here. />

6.2 Test Report

<TIP: Describe how you test your project. Place a test report here. If you use continuous integration and deployment (CI/CD) tools, describe your CI/CD method here. />

Chapter 7

Conclusion and Discussion

7.1 Summary of Achievements

<TIP: Summarize what your project accomplished. Refer back to your problem statement from Chapter 1 and explain the degree to which your solution addressed it. Highlight the most significant features or results delivered. />

7.2 Known Limitations

<TIP: Describe the current limitations of your solution honestly. These may include technical constraints (e.g., scalability, performance), scope limitations (features not yet implemented), data limitations (e.g., small training sets, biased data), or usability issues discovered during testing. />

7.3 Future Work

<TIP: Propose concrete improvements or extensions that could be made to this project given more time or resources. These should follow logically from your limitations. For example, if your model was trained on limited data, future work might include collecting a larger dataset. Avoid vague suggestions like "improve the UI"; instead, be specific about what would be changed and why. />

Reference

Bibliography

[1] Overleaf, “Learn latex in 30 minutes,” https://www.overleaf.com/learn/latex/Learn_LaTeX_in_30_minutes.

Appendix A

Appendix A: Example

<TIP: Put additional or supplementary information/data/figures in appendices. />

Appendix B

Appendix B: About LaTeX

LaTeX (stylized as LaTeX) is a software system for typesetting documents. LaTeX markup describes the content and layout of the document, as opposed to the formatted text found in WYSIWYG word processors like Google Docs, LibreOffice Writer, and Microsoft Word. The writer uses markup tagging conventions to define the general structure of a document, to stylize text throughout a document (such as bold and italics), and to add citations and cross-references.

LaTeX is widely used in academia for the communication and publication of scientific documents and technical note-taking in many fields, owing partially to its support for complex mathematical notation. It also has a prominent role in the preparation and publication of books and articles that contain complex multilingual materials, such as Arabic and Greek.

Overleaf has also provided a 30-minute guide on how you can get started on using LaTeX. [**?**]